Computer Science

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Overview Programs Courses

CODE	NAME	DESCRIPTION
CMPSC1	Seminar on the Field of Computer Science	Overviews the potential of, and opportunities available from, the field of computer science. Topics include an overview of how computers work and the interesting ways in which computers can be applied to solve important and high-impact technological,...
CMPSC100	Introduction to Teaching Methods in Computer Science	Designed to train outstanding undergraduates for learning assistant positions in CS courses. Lecture/discussion surveys current research and best practices in CS pedagogy including student development theories, different pedagogical techniques, and m...
CMPSC11	Programming Language Laboratory	A self-paced course to allow a student who already possesses a working knowledge of at least one programming language an opportunity to learn other languages of interest. Each section studies a different programming language.
CMPSC110	Introduction to Research in Computer Science	Defining a CS research problem, finding and reading technical papers, oral communication, technical writing, and independent learning. Course participants work in teams as they apprentice with a CS research group to propose an original research probl...

CODE	NAME	DESCRIPTION
CMPSC111	Introduction to Computational Science	Introduction to the numerical algorithms that form the foundations of data science, machine learning, and computational science and engineering. Matrix computation, linear equation systems, eigenvalue and singular value decompositions, numerical opti...
CMPSC11AD	Programming Language Laboratory: ADA	A self-paced course to allow a student who already possesses working knowledge of at least one programming language an opportunity to learn other languages of interest. Each section studies a different language.
CMPSC11AP	Programming Language Laboratory: APL	A self-paced course to allow a student who already possesses working knowledge of at least one programming language an opportunity to learn other languages of interest. Each section studies a different language.
CMPSC11C	Programming Language Laboratory	A self-paced course to allow a student who already possesses a working knowledge of at least one programming language an opportunity to learn other languages of interest. Each section studies a different language.
CMPSC11CO	Programming Language Laboratory: COBOL	A self-paced course to allow a student who already possesses a working knowledge of at least one programming language an opportunity to learn other languages of interest. Each section studies a different language.
CMPSC11FO	Programming Language Laboratory: Fortran	A self-paced course to allow a student who already possesses a working knowledge of at least one programming language an opportunity to learn other languages of interest. Each section studies a different language.
CMPSC11JA	Programming Language Laboratory	A self-paced course to allow a student who already possesses a working knowledge of at least one programming language an opportuity to learn otherlanguages of interest. Each section studies a different language. Sections are: 11PA-Pascal, 11FO-Fortra...

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CMPSC11LI	Programming Language Laboratory: LISP	A self-paced course to allow a studen who already possesses a working knowledge of at least one programming language an opportunity to learn other languages of interest. Each section studies a different language.
CMPSC11PA	Programming Language Laboratory: Pascal	A self-paced course to allow a student who already possesses working knowledge of at least one programming language an opportunity to learn other languages of interest. Each section studies a different language.
CMPSC11PL	Programming Language Laboratory: PL/1	A self-paced course to allow a studen who already possesses a working knowledge of at least one programming language an opportunity to learn other languages of interest. Each section studies a different language.
CMPSC11SN	Programming Language Laboratory: SNOBOL	A self-paced course to allow a student who already possesses working knowledge of at least one programming language an opportunity to learn other languages of interest. Each section studies a different language.
CMPSC130A	Data Structures and Algorithms	Data structures and applications with proofs of correctness and analysis. Hash Tables, Priority Queues (heaps); Balanced Search Trees. Graph Traversal techniques and their applications.
CMPSC130B	Data Structures and Algorithms II	Design and analysis of computer algorithms. Correctness proofs and solution of recurrence relations. Design techniques; divide and conquer, greedy strategies, dynamic programming. Applications of techniques to problems from several disciplines. NP -...
CMPSC132	Algorithms Engineering	Applying theoretical knowledge about algorithms, data structures, and complexity to performing the complete process of analyzing a computational problem, estimating the running time of suggested solutions, and implementing an effective and correct pr...

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CMPSC134	Randomized Algorithms	Covers advanced algorithmic ideas, focusing on randomized algorithms. Explores how randomization can be used to make algorithms simpler and more efficient, and to understand the probabilistic analysis of algorithms.
CMPSC138	Automata and Formal Languages	Formal languages; finite automata and regular expressions; properties of regular languages; pushdown automata and context-free grammars; properties of context-free languages; introduction to Turing machines and computability.
CMPSC140	Parallel Scientific Computing	Fundamentals of high performance computing and parallel algorithm design for numerical computation. Topics include parallel architectures and clusters, parallel programming with message-passing libraries and threads, program parallelization methodolo...
CMPSC148	Computer Science Project	Team-based project development. Topics include software engineering and professional development practices, interface design, advanced library support; techniques for team oriented design and development, testing and test driven development, and soft...
CMPSC153A	Hardware/Software Interface	Design and coding of embedded programs. Real-time constraints and management of resources. Practical I/O interfaces and protocols, polling, interrupt maintenance, atomic code, blocking and non-blocking tasks. Static and dynamic task scheduling under...
CMPSC154	Computer Architecture	Introduction to the architecture of computer systems. Topics include: central processing units, memory systems, channels and controllers, peripheral devices, interrupt systems, software versus hardware tradeoffs.
CMPSC156	Advanced Applications Programming	Advanced application programming using a high-level, virtual-machine-based language. Topics include generic programming, exception handling, automatic memory management, and

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		application development, management, and maintenance tools, third-party libr...
CMPSC16	Problem Solving with Computers I	Fundamental building blocks for solving problems using computers. Topics include basic computer organization and programming constructs: memory CPU, binary arithmetic, variables, expressions, statements, conditionals, iteration, functions, parameters...
CMPSC160	Translation of Programming Languages	Study of the structure of compilers. Topics include: lexical analysis; syntax analysis including LL and LR parsers; type checking; run-time environments; intermediate code generation; and compiler-construction tools.
CMPSC162	Programming Languages	Concepts of programming languages: scopes, parameter passing, storage management; control flow, exception handling; encapsulation and modularization mechanisms; reusability through genericity and inheritance; type systems; programming paradigms (impe...
CMPSC165A	Artificial Intelligence	Introduction to the field of artificial intelligence, which seeks to understand and build intelligent computational systems. Topics include intelligent agents, problem solving and heuristic search, knowledge representation and reasoning, uncertainty...
CMPSC165B	Machine Learning	Covers the most important techniques of machine learning (ML) and includes discussions of: well-posed learning problems; artificial neural networks; concept learning and general to specific ordering; decision tree learning; genetic algorithms; Bayesi...
CMPSC170	Operating Systems	Basic concepts of operating systems. The notion of a process; interprocess communication and synchronization; input-output, file systems, memory management.
CMPSC171	Distributed Systems	Distributed systems architecture, distributed programming, network of computers, message passing, remote procedure calls, group

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		communication, naming and membership problems, asynchrony, logical time, consistency, fault-tolerance, and recovery.
CMPSC172	Software Engineering	Software engineering is concerned with long-term, large-scale programming projects. Software management, cost estimates, problem specification and analysis, system design techniques, system testing and performance evaluation, and system maintenance....
CMPSC174A	Fundamentals of Database Systems	Database system architectures, relational data model, relational algebra, relational calculus, SQL, QBE, query processing, integrity constraints (key constraints, referential integrity), database design, ER and object- oriented data model, functional...
CMPSC174B	Design and Implementation Techniques of Database Systems	Queries and processing, optimizer, cost models, execution plans, rewriting rules, access methods, spatial indexing; transactions, ACID properties, concurrency control, serializability, two-phase locking, timestamping, logging,checkpointing, transacti...
CMPSC174N	Introduction to Databases	ER/object-oriented data modeling, relational data model, database design, query languages (algebra, calculus, SQL, QBE), updates and views, query processing, integrity constraints (key constraints, referential integrity), functional dependence, Boyce...
CMPSC176A	Introduction to Computer Communication Networks	Basic concepts in networking, the OSI model, error detection codes, flow control, routing, medium access control, and high-speed networks.
CMPSC176B	Network Computing	Focus on networking and web technologies used in the Internet. The class covers socket programming and web-based techniques that are used to build distributed applications.
CMPSC176C	Advanced Topics in Internet Computing	General overview of wireless and mobile networking, multimedia, security, multicast, quality of service, IPv6, and web caching. During

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		the second half of the course, one or more of the above are studied in greater detail.
CMPSC177	Computer Security	Introduction to the basics of computer security and privacy. Analysis of technical difficulties of producing secure computer information systems that provide guaranteed controlled sharing. Examination and critique of current systems, methods, certifi...
CMPSC178	Introduction to Cryptography	An introduction to the basic concepts and techniques of cryptography and cryptanalysis. Topics include: The Shannon Theory, classical systems, the enigma machine, the data encryption standard, public key systems, digital signatures, file security.
CMPSC180	Computer Graphics	Overview of OpenGL graphics standard, OpenGL statemachine, other 3D graphics libraries, 3D graphicspipeline, 3D transformations and clipping, colormodel, shading model, shadow algorithms, texturing,curves and curved surfaces, graphics hardware,intera...
CMPSC181	Introduction to Computer Vision	Overview of computer vision problems and techniques for analyzing the content of images and video. Topics include image formation, edge detection, image segmentation, pattern recognition, texture analysis, optical flow, stereo vision, shape represent...
CMPSC182	Multimedia Computing	Introduction to multimedia and applications. Topics include streaming media, conferencing, webcasting, digital libraries, multimedia system architectures, standards (including JPEG and MPEG), and multimedia storage and retrieval. A key emphasis is on...
CMPSC184	Mobile Application Development	An introduction to programming mobile computing devices. Students will learn about and study the shift in software development from desktop to mobile device applications. Topics will include software engineering and design practices, advances in prog...

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CMPSC185	Human-Computer Interaction	The study of human-computer interaction enables system architects to design useful, efficient, and enjoyable computer interfaces. This course teaches the theory, design guidelines, programming practices, and evaluation procedures behind effective hum...
CMPSC186	Internet of Things (IoT) Systems	The Internet of Things (IoT) embeds ordinary physical objects in our environment with digital intelligence -- via sensing, control, communications, and computing capabilities. As such, IoT advances are likely key drivers of the next computing revolut...
CMPSC188	Introduction to Natural Language Processing	This NLP course explores AI's capability to understand human language, focusing on machine learning algorithms and large language models. It covers language modeling, sentiment analysis, tagging, and machine translation, discussing lexical, syntactic...
CMPSC189A	Senior Computer Systems Project	This course is the first in the 2-course Computer Science Capstone sequence. Student teams learn about modern software design and engineering techniques and work together to use them and their past educational experiences at UCSB to analyze, specify,...
CMPSC189B	Senior Computer Systems Project	This course is the second in the 2-course Computer Science Capstone sequence. Student teams continue their CMPSC 189A effort by evolving their specifications, design, and prototypes of advanced computer science solutions and systems. The students lea...
CMPSC190	Special Topics in Computer Science	Provides for the study of topics of current interest in computer science: A. Foundations; B. Software Systems; C. Programming Languages and Software Engineering; D. Information Management; E. Architecture; F. Networking; G. Security; H. Scientific Co...
CMPSC190A	Special Topics in Computer Science:	These variable unit courses provide for the study of topics of current interest in computer science.

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	Foundations	
CMPSC190B	Special Topics in Computer Science: Software Systems	These variable unit courses provide for the study of topics of current interest incomputer science.
CMPSC190C	Special Topics in Computer Science: Programming Languages and Software Engineering	These variable unit courses provide for the study of topics of current interest in computer science.
CMPSC190D	Special Topics in Computer Science: Information Management	These variable unit courses provide for the study of topics of current interest in computer science.
CMPSC190DA	Special Topics in Computer Science	Provides for the study of topics of current interest in computer science: A. Foundations; B. Software Systems; C. Programming Languages and Software Engineering; D. Information Management; E. Architecture; F. Networking; G. Security; H. Scientific Co...
CMPSC190DB	Special Topics in Computer Science	Provides for the study of topics of current interest in computer science: A. Foundations; B. Software Systems; C. Programming Languages and Software Engineering; D. Information Management; E. Architecture; F. Networking; G. Security; H. Scientific Co...
CMPSC190DC	Special Topics in Computer Science	Provides for the study of topics of current interest in computer science: A. Foundations; B. Software Systems; C. Programming Languages and Software Engineering; D. Information Management; E. Architecture; F. Networking; G. Security; H. Scientific Co...
CMPSC190DD	Special Topics in Computer Science	Provides for the study of topics of current interest in computer science: A. Foundations; B. Software Systems; C. Programming Languages and Software Engineering; D. Information Management; E. Architecture; F. Networking; G. Security; H. Scientific Co...

CODE	NAME	DESCRIPTION
CMPSC190DE	Special Topics in Computer Science	Provides for the study of topics of current interest in computer science: A. Foundations; B. Software Systems; C. Programming Languages and Software Engineering; D. Information Management; E. Architecture; F. Networking; G. Security; H. Scientific Co...
CMPSC190DF	Special Topics in Computer Science	Provides for the study of topics of current interest in computer science: A. Foundations; B. Software Systems; C. Programming Languages and Software Engineering; D. Information Management; E. Architecture; F. Networking; G. Security; H. Scientific Co...
CMPSC190E	Special Topics in Computer Science: Architectures	These variable unit courses provide for the study of topics of current interest in computer science.
CMPSC190F	Special Topics in Computer Science: Networking	These variable unit coureses provide for the study of topics of current interest in computer science.
CMPSC190G	Special Topics in Computer Science: Security	These variable unit courses provide for the study of topics of current interest in computer science.
CMPSC190H	Special Topics in Computer Science: Scientific Computing	These variable unit courses provide for the study of topics of current interest in computer science.
CMPSC190I	Special Topics in Computer Science: Intelligent and Interactive Systems	These variable unit courses provide for the study of topics of current interest in computer science.
CMPSC190J	Special Topics in Computer Science: General	These variable unit courses provide for the study of topics of current interest in computer science.
CMPSC190K	Special Topics in Computer Science: Computer Systems Modeling and Analysis	These variable unit courses provide for the study of topics of current interest in computer science.

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CMPSC190N	Special Topics in Computer Science: General	Provides for the study of topics of current interest in computer science: A. Foundations; B. Software Systems; C. Programming Languages and Software Engineering; D. Information Management; E. Architecture; F. Networking; G. Security; H. Scientific Co...
CMPSC192	Projects in Computer Science	Projects in computer science for advanced undergraduate students.
CMPSC193	Internship in Industry	Special projects for selected students. Offered in conjunction with selected industrial and research firms and under direct faculty supervision. Written proposal and final report required.
CMPSC196	Undergraduate Research	Research opportunities for undergraduate students. Students will be expected to giveregular oral presentations, actively participate in a weekly seminar, and prepare at least one written report on their research.
CMPSC196B	Undergraduate Research	Advanced research for undergraduate students, by petition after completing a minimum of 4 units of CMPSC 196 for a letter grade. The student will propose a specific research project and make a public presentation of final results. Evaluation and grad...
CMPSC199	Independent Studies in Computer Science	Independent studies in computer science for advanced students.
CMPSC2	Computational Thinking for Scientists	Introduction to the central concepts of computer science as they apply to a wide variety of human endeavors, including natural and physical sciences. Topics that interrelate central ideas from algorithms, performance and complexity, data structures,...
CMPSC209	Logic and Applications in Computer Science	Propositional logic, first order logic, completeness, compactness, incompleteness, undecidability; selected topics from finite model theory, theorem proving, logic programming, program verification, databases, computational complexity.

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CMPSC211A	Matrix Analysis and Computation	Graduate level-matrix theory with introduction to matrix computations. SVD's, pseudoinverses, variational characterization of eigenvalues, perturbation theory, direct and iterative methods for matrix computations.
CMPSC211B	Numerical Simulation	Linear multistep methods and Runge-Kutta methods for ordinary differential equations: stability, order and convergence. Stiffness. Differential algebraic equations. Numerical solution of boundary value problems.
CMPSC211C	Numerical Solution of Partial Differential Equations--Finite Difference Methods	Finite difference methods for hyperbolic, parabolic and elliptic PDE's, with application to problems in science and engineering. Convergence, consistency, order and stability offinite difference methods. Dissipation and dispersion. Finite volume meth...
CMPSC211D	Numerical Solution of Partial Differential Equations--Finite Element Methods	Weighted residual and finite element methods for the solution of hyperbolic, parabolic and elliptical partial differential equations, with application to problems in science and engineering. Error Estimates. Standard and discontinuous Galerkin method...
CMPSC216	Level Set Methods	Mathematical description of the level set method and design of the numerical methods used in its implementations (ENO-WENO, Godunov, Lax-Friedrich, etc.). Introduction to the Ghost Fluid Method. Applications in CFD, Materials Sciences, Computer Visio...
CMPSC219	Sparse Matrix Algorithms	Theory, implementation, and applications of sparse matrix algorithms. Direct, iterative, preconditioned, and multilevel methods for solving large, sparse systems of linear equations. Tools include graph theory, linear algebra, and computer architectu...
CMPSC220	Theory of Computation and Complexity	Topics include: models of computation; time and space complexity classes (e.g., P, NP, CO-NP, and Pspace), efficient reducibilities, complete problems lower bounds; the polynomial

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		hierarchy.
CMPSC225	Information Theory	Entropy, mutual information, and Shannon's coding theorems; lossless source coding, Huffman, Shannon-Fano-Elias, and arithmetic codes; Channel capacity; rate-distortion theory, and lossy source coding; source-channel coding; algorithmic complexity an...
CMPSC230	Approximations, NP-Completeness and Algorithms	Epsilon approximations, PTAS and FPTAS. Techniques for the design of approximation algrorithms. P, NP, NP-complete problems, polynomial transformations, Turing reductions, strong NP-completeness, NP-hardness and inapproximability results. Topics in a...
CMPSC231	Topics in Combinatorial Algorithms	Advanced topics in algorithm design, including network flows, matchings in graphs, linear and integer programming.
CMPSC235	Computational Geometry	Algorithms and lower bound techniques in computational geometry; decision tree models of computation; geometric searching; point location and range search; convex hull and maxima of a point set; proximity algorithms; geometric intersections.
CMPSC24	Problem Solving with Computers II	Intermediate building blocks for solving problems using computers. Topics include intermediate object-oriented programming, data structures, object-oriented design, algorithms for manipulating these data structures and their run-time analyses. Data s...
CMPSC240A	Applied Parallel Computing	Interdisciplinary introduction to applied parallel computing on modern supercomputers. Topics include applications-oriented architectural issues, MPI, parallel MATLAB, and parallel numerical algorithms. A course project emphasizes understanding the r...
CMPSC24A	Problem Solving with Computers II	Intermediate building blocks for solving problems using computers. Topics include data structures, object-oriented design and development, algorithms for manipulating these

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		data structures and their runtime analyses. Data structures introduced includ...
CMPSC254	Advanced Computer Architecture	Advanced instruction set architectures, pipelining, dynamic scheduling, branch prediction, superscalar issue, out-of-order execution, memory-hierarchy design, advanced cache architectures and prefetching. Several real designs are dissected and simula...
CMPSC260	Advanced Topics in Program Analysis	Theoretical aspects of translation. Topics include: data flow analysis; control flow analysis; interprocedural analysis; optimization; type systems.
CMPSC263	Runtime Systems	Topics central to modern programming languages and their implementation: garbage collection; memory system performance; characteristics and optimization of object-oriented languages; type systems and type inference; run-time compilation.
CMPSC266	Formal Specification and Verification	Introduction to existing specification and verification systems, and the underlying theory and techniques of verifying the correctness of algorithms with respect to specifications. This subject can be considered as the combination of specification an...
CMPSC267	Automated Verification	Covers automated verification algorithms and tools. Topics include: temporal logics, fixpoint characterizations of temporal properties, model checking, symbolic verification, explicit-state verification, verification using automated theorem provers,...
CMPSC270	Operating Systems	Develops an understanding of operating systems and operating systems research at both a conceptual level and from an engineering perspective. Fundamental design principles, culled primarily from the research literature, motivate an analysis of previo...
CMPSC271	Advanced Topics in Distributed Systems	Covers the fundamental problems in distributed systems and the various tools used to solve them.

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		Of primary interest is the issue of fault-tolerance. Topics include event ordering, clocks, global states, agreement, fault-tolerance, and peer-to-peer s...
CMPSC272	Software Engineering	Principles of software engineering disciplines emphasizing requirements analysis, specification design, coding, testing and correctness proofs, maintenance, and management. Students use a number of software engineering tools.
CMPSC273	Data and Knowledge Bases	The focus is on the study of relational and post-relational data models and their query languages of different styles (algebraic, calculus, and deductive): complexity, expressive power, optimization, and database design.
CMPSC274	Advanced Topics in Database Systems	Topics include: data models, semantics, data integrity, database design, serializability theory, concurrency control, recovery, distributed databases.
CMPSC276	Advanced Topics in Networking	Focuses on advanced topics in networking. Topics may include, but are not limited to: Internet analysis, routing techniques, multimedia, approaches for network performance enhancements, and communication over new technologies.
CMPSC279	Advanced Topics in Security	Security analysis of networked systems. Analysis of techniques for network scanning, spoofing, hijacking, and denial-of-service attacks. Vulnerability analysis of applications and web-based systems. Intrusion detection and prevention techniques.
CMPSC280	Computer Graphics	Special topics in computer graphics including: curves and curved surfaces; visual perception of colors and color models; shading models; shadow generation; texture mapping; solid textures; stereographics; helmet-mounteddisplay; graphics hardware/arch...
CMPSC281	Advanced Topics in Computer Vision	Topics in Computer Vision, including imaging and multi-view geometry; image and video

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		segmentation; object detection, localization, and recognition; video activity recognition; machine learning models and methods applied to computer vision.
CMPSC281B	Advanced Topics in Computer Vision	Advanced topics in computer vision: image sequence analysis, spatio-temporal filtering, camera calibration and hand-eye coordination, robot navigation, shape representation, physically-based modeling, regularization theory, multi-sensory fusion, biol...
CMPSC284	Mobile Computing	Focuses on mobile computing. Topics may include, but are not limited to: mobile network characteristics, types of mobile networks, challenges and sollutions in mobile computing, and power conservation techniques.
CMPSC285	Advanced Image Synthesis	Theory and design of rendering systems for high-end image synthesis and modeling physically-based light transport. Topics include REYES-style architectures, Monte Carlo algorithms, variance reduction methods, path tracing, photon mapping, reflection...
CMPSC291	Special Topics in Computer Science, Applications	These courses provide for the study of topics of current interest in computer science applications. Topics are coded as follows: A. Applications, General; D. Database and Information Systems; G. Security and Cryptography; H. Computational Science and...
CMPSC291A	Special Topics in Computer Science, Applications, General	These courses provide for the study of topics of current interest in computer science applications.
CMPSC291D	Special Topics in Computer Science, Applications, Database and Information Systems	These courses provide for the study of topics of current interest in computer science applications.
CMPSC291G	Special Topics in Computer Science, Applications, Security and Cryptography	These courses provide for the study of topics of current interest in computer science applications.

CODE	NAME	DESCRIPTION
CMPSC291H	Special Topics in Computer Science, Applications, Computational Science and Engineering	These courses provide for the study of topics of current interest in computer science applications.
CMPSC291I	Special Topics in Computer Science, Applications, Visual Computing and Interaction	These courses provide for the study of topics of current interest in computer science applications.
CMPSC291K	Special Topics in Computer Science, Applications	These courses provide for the study of topics of current interest in computer science applications. Topics are coded as follows: A. Applications, General; D. Database and Information Systems; G. Security and Cryptography; H. Computational Science and...
CMPSC292	Special Topics in Computer Science, Foundations	These courses provide for the study of topics of current interest in computer science foundations. Topics are coded as follows: C, Programming Languages and Software Engineering; F. Foundations, General; G. Security and Cryptography; H. Computational...
CMPSC292A	Special Topics in Computer Science, Foundations, Algorithms	These courses provide for the study of topics of current interest in computer science foundations.
CMPSC292C	Special Topics in Computer Science, Foundations, Programming Languages and Software Engineering	These courses provide for the study of topics of current interest in computer science foundations.
CMPSC292F	Special Topics in Computer Science, Foundations, General	These courses provide for the study of topics of current interest in computer science foundations.
CMPSC292G	Special Topics in Computer Science, Foundations	These courses provide for the study of topics of current interest in computer science foundations. Topics are coded as follows: C, Programming Languages and Software Engineering; F.

CODE	NAME	DESCRIPTION
		Foundations, General; G. Security and Cryptography; H. Computational...
CMPSC292H	Special Topics in Computer Science, Foundations, Computational Science and Engineering	These courses provide for the study of topics of current interest in computer science foundations.
CMPSC293	Special Topics in Computer Science, Systems	These courses provide for the study of topics of current interest in computer science systems. Topics are coded as follows: B. Operating Systems and Distributed Systems; C, Programming Languages and Software Engineering; D. Database and Information S...
CMPSC293A	Special Topics in Computer Science, Systems	These courses provide for the study of topics of current interest in computer science systems. Topics are coded as follows: B. Operating Systems and Distributed Systems; C, Programming Languages and Software Engineering; D. Database and Information S...
CMPSC293B	Special Topics in Computer Science, Systems	These courses provide for the study of topics of current interest in computer science systems. Topics are coded as follows: B. Operating Systems and Distributed Systems; C, Programming Languages and Software Engineering; D. Database and Information S...
CMPSC293C	Special Topics in Computer Science, Systems, Programming Languages and Software Engineering	These courses provide for the study of topics of current interest in computer science systems.
CMPSC293D	Special Topics in Computer Science, Systems, Database and Information Systems	These courses provide for the study of topics of current interest in computer science systems.
CMPSC293E	Special Topics in Computer Science, Systems, Computer Architecture and	These courses provide for the study of topics of current interest in computer science systems.

CODE	NAME	DESCRIPTION
	Embedded Systems	
CMPSC293G	Special Topics in Computer Science, Systems, Security and Cryptography	These courses provide for the study of topics of current interest in computer science systems.
CMPSC293K	Special Topics in Computer Science, Systems, Machine Learning and Data Mining	These courses provide for the study of topics of current interest in computer science systems.
CMPSC293N	Special Topics in Computer Science, Systems	These courses provide for the study of topics of current interest in computer science systems. Topics are coded as follows: B. Operating Systems and Distributed Systems; C, Programming Languages and Software Engineering; D. Database and Information S...
CMPSC293S	Special Topics in Computer Science, Systems, General	These courses provide for the study of topics of current interest in computer science systems.
CMPSC32	Object Oriented Design and Implementation	Advanced topics in object-oriented computing. Topics include encapsulation, data hiding, inheritance, polymorphism, compilation, linking and loading, memory management, and debugging; recent advances in design and development tools, practices, librar...
CMPSC4	Computer Science Boot Camp	An introduction to computational thinking, computing, data management, and problem solving using computers, for non-majors. Topics include coding basics, representing code and data using a computer, and applications of computing that are important to...
CMPSC40	Foundations of Computer Science	Introduction to the theoretical underpinnings of computer science. Topics include propositional predicate logic, set theory, functions and relations, counting, mathematical induction and recursion (generating functions).

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CMPSC501	Techniques of Computer Science Teaching	An initial 1-2 day workshop on teaching techniques followed by a weekly seminar. Course emphasizes teaching skills, practical experience, and communication skills.
CMPSC502	Teaching of Computer Science	Procedures and techniques for teaching computer science gained through actual teaching of lecture courses, leading discussion sections, and/or teaching laboratories. Meetings will be held as needed to discuss problems,methods, and procedures.
CMPSC592	Group Studies: Controls, Dynamical Systems, and Computation	A series of weekly lectures given by university staff and outside experts in the fields of control systems, dynamical systems, and computation.
CMPSC593	Computer Science Graduate Tutorial	Individual studies exploring topics in computer science with a faculty advisor.
CMPSC594	Seminar in Computer Science	A seminar course offered on an irregular basis. Provides an in-depth discussion of advanced topics of general interest and broadens the scope ofknowledge in computer science.
CMPSC595	Group Studies in Computer Science	Special seminars focusing on topics of interest to faculty and graduate students. These seminars provide critical review of research in various areas of computer science: A. Foundations; B. Software Systems; C. Programming Languages and Software Engi...
CMPSC595A	Group Studies in Computer Science: Foundations	Special seminars focusing on topics of interest to faculty and graduate students. These seminars provide critical review of research in various areas of computer science.
CMPSC595B	Group Studies in Computer Science: Software Systems	Special seminars focusing on topics of interest to faculty and graduate students. These seminars provide critical review of research in various areas of computer science.

CODE	NAME	DESCRIPTION
CMPSC595C	Group Studies in Computer Science: Programming Languages and Software Engineering	Special seminars focusing on topics of interest to faculty and graduate students. These seminars provide critical review of research in various areas of computer science.
CMPSC595D	Group Studies in Computer Science	Special seminars focusing on topics of interest to faculty and graduate students. These seminars provide critical review of research in various areas of computer science.
CMPSC595E	Group Studies in Computer Science: Architecture	Special seminars focusing on topics of interest to faculty and graduate students. These seminars provide critical review of research in various areas of computer science.
CMPSC595F	Group Studies in Computer Science: Networking	Special seminars focusing on topics of interest to faculty and graduate students. These seminars provide critical review of research in various areas of computer science.
CMPSC595G	Group Studies in Computer Science: Security	Special seminars focusing on topics of interest to faculty and graduate students. These seminars provide critical review of research in various areas of computer science.
CMPSC595H	Group Studies in Computer Science: Scientific Computing	Special seminars focusing on topics of interest to faculty and graduate students. These seminars provide critical review of research in various areas of computer science.
CMPSC595I	Group Studies in Computer Science: Intelligent and Interactive Systems	Special seminars focusing on topics of interest to faculty and graduate students. These seminars provide critical review of research in various areas of computer science.
CMPSC595J	Group Studies in Computer Science: General	Instruction in these variable unit courses may be carried out by lecture, or by laboratory, or by a combination of the two. These courses provide critical review of research in various areas of computer science.
CMPSC595K	Group Studies in Computer Science: Computer Systems	Instruction in these variable unit courses may be carried out by lecture, or by laboratory, or by a combination of the two. These courses provide

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	Modeling and Analysis	critical review of research in various areas of computer science.
CMPSC595L	Group Studies in Computer Science	Instruction in these variable unit course may be carried out by lecture, or by laboratory, or by a combination of the two. These courses provide critical review of research in various areas of computer science.
CMPSC595N	Group Studies in Computer Science: Faculty Research Seminar	Special seminars focusing on topics of interest to faculty and graduate students. These seminars provide critical review of research in various areas of computer science.
CMPSC595R	Group Studies in Computer Science	Instruction in these variable unit course may be carried out by lecture, orby laboratory, or by a combination of the two. These courses provide critical review of research in various areas of computer science. See gcmps290AA-ZZ for list of topics. A:...
CMPSC596	Directed Research	Research, either experimental or theoretical, may be undertaken by qualified graduate students under the direction of a faculty member.
CMPSC597	Individual Studies for Master of Science Comprehensive Examinations and Ph.D Examinations	Individual studies for M.S. comprehensive examination and Ph.D.
CMPSC598	Master's Thesis Research and Preparation	For research underlying the thesis and writing of the thesis.
CMPSC599	Ph.D. Dissertation Research and Preparation	Research and preparation of dissertation.
CMPSC5A	Introduction to Data Science 1	Introduction to data science methods and Python programming language for students with little to no experience in the subjects. Topics include foundational programming concepts, problem-solving strategies and code design, and data science concepts as...

CODE	NAME	DESCRIPTION
CMPSC5B	Introduction to Data Science 2	Students explore the data science lifecycle, including question formulation, data collection and cleaning, exploratory data analysis and visualization, statistical inference and prediction, and decision-making. The course focus is on transforming and...
CMPSC64	Computer Organization and Logic Design	Assembly language programming and advanced computer organization; Digital logic design topics including gates, combinational circuits, flip-flops, and the design and analysis of sequential circuits.
CMPSC8	Introduction to Computer Science	Introduction to computer program development for students with little to no programming experience. Basic programming concepts, variables and expressions, data and control structures, algorithms, debugging, program design, and documentation.
CMPSC9	Intermediate Python Programming	Intermediate topics in Computer Science using the Python programming language. Topics include object oriented programming, runtime analysis, data structures, and software testing methodologies.
CMPSC90	Special Topics in Computer Science	Provides for the study of topics of current interest in computer science: A. Foundations; B. Software Systems; C. Programming Languages and Software Engineering; D. Information Management; E. Architecture; F. Networking; G. Security; H. Scientific Co...
CMPSC90DA	Special Topics in Computer Science	Provides for the study of topics of current interest in computer science: A. Foundations; B. Software Systems; C. Programming Languages and Software Engineering; D. Information Management; E. Architecture; F. Networking; G. Security; H. Scientific Co...
CMPSC90DB	Special Topics in Computer Science	Provides for the study of topics of current interest in computer science: A. Foundations; B. Software Systems; C. Programming Languages and Software Engineering; D. Information

CODE	NAME	DESCRIPTION
		Management; E. Architecture; F. Networking; G. Security; H. Scientific Co...
CMPSC96	Undergraduate Research	Research opportunities for undergraduate students. Students will be expected to give regular oral presentations, actively participate in a weekly seminar, and prepare at least one written report on their research.
CMPSC99	Independent Studies in Computer Science	Independent studies in computer science for advanced students.
CMPSCW8	Introduction to Computer Science	Introduction to computer program development for students with little to no programming experience. Basic programming concepts, variables and expressions, data and control structures, algorithms, debugging, program design, and documentation.
CMPSCW9	Intermediate Python Programming	Intermediate topics in Computer Science using the Python programming language. Topics include object oriented programming, runtime analysis, data structures, and software testing methodologies.

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